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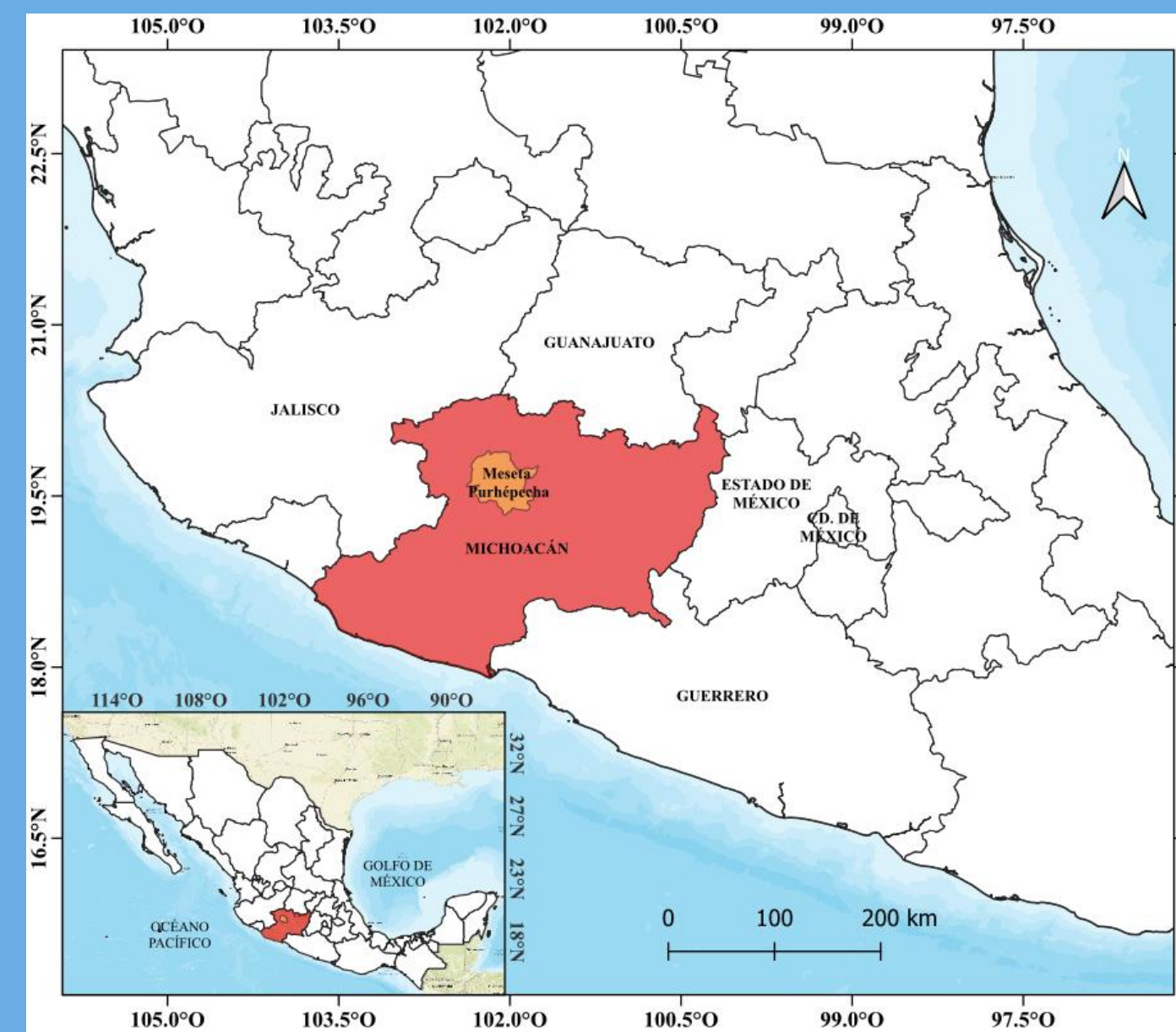
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Fire and Territory: An Exploratory Analysis of Cultural Fire Regimes in Indigenous Communities of Michoacán

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Introduction

The use of fire by Indigenous communities is a socio-ecological phenomenon in which practices, norms, and knowledge interact with biodiversity and landscape regeneration (Long et al., 2021; Vázquez-Varela et al., 2022). These socially regulated and intentional practices encompass cultural, agricultural, forestry, and ritual burnings, characterized by their frequency, scale, and community-based control mechanisms (Martínez-Torres et al., 2016; Long et al., 2021).

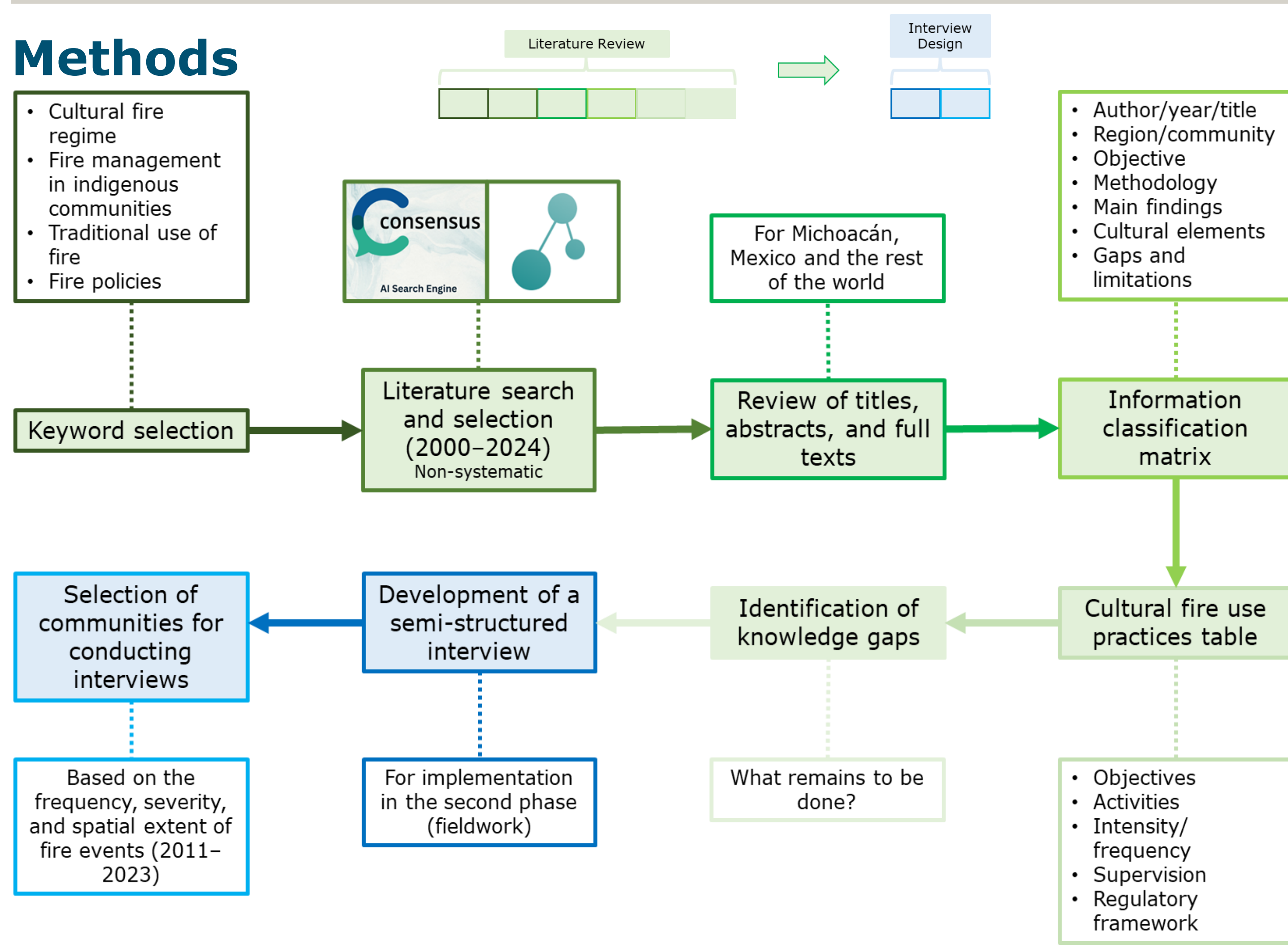
At the international level, Indigenous fire management has been documented to promote biodiversity, cultural cohesion, and resilience to extreme wildfire events (Bardsley et al., 2019; Hoffman, 2021; Ponce-Calderón et al., 2022; Smith et al., 2022, 2024).

In Mexico, these strategies integrate local ecological knowledge and collective agreements, although they often come into tension with formal regulatory frameworks (Rodríguez-Trejo et al., 2011; Fulé et al., 2011; Ponce-Calderón et al., 2022). However, knowledge gaps remain regarding who makes decisions, how these practices are organized, and which norms guide them—particularly in the Meseta Purépecha.

Research question

What components, perceptions, and legal frameworks are associated with indigenous cultural fire regimes, and how have these been examined in the scientific and documentary literature both within Mexico and internationally?

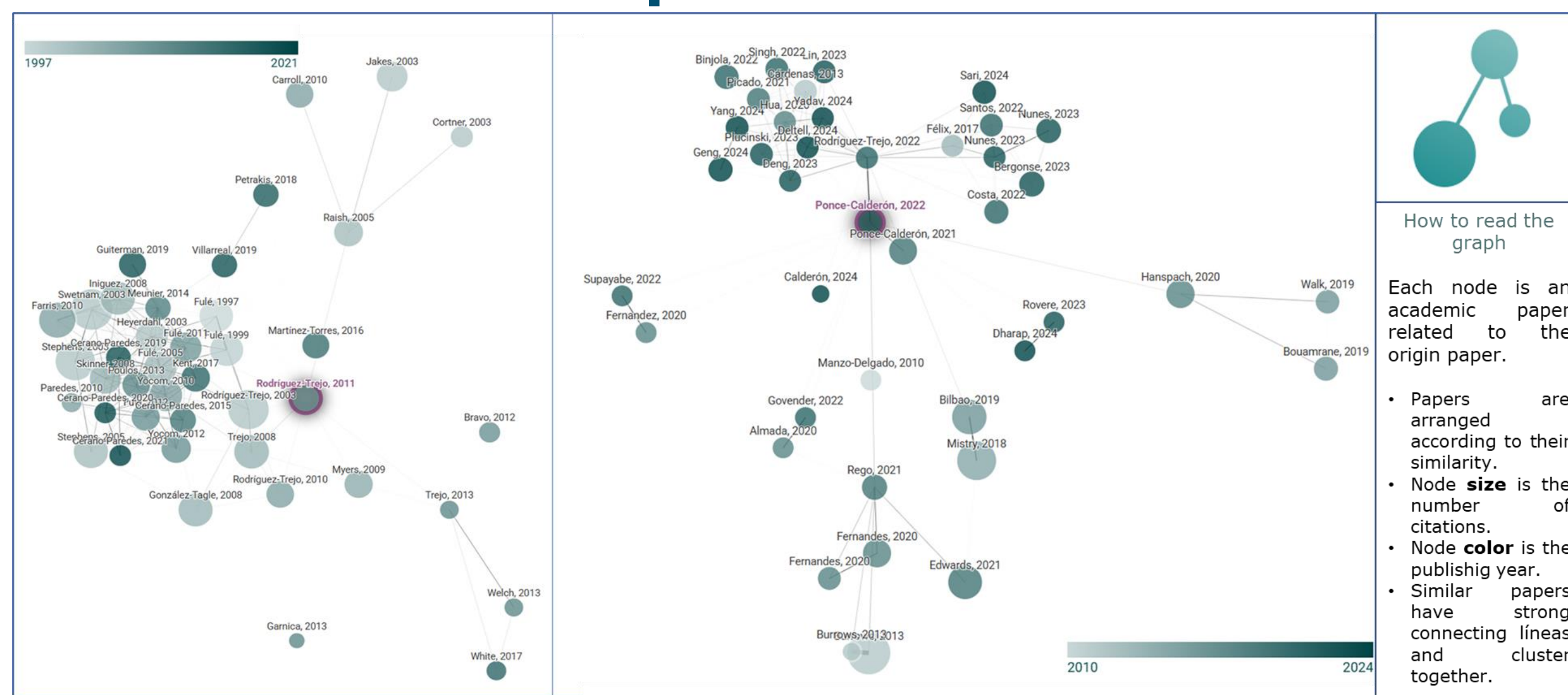
Methods



References

- *Bardsley, D. K., Prowse, T. A., & Siegfriedt, C. (2019). Seeking knowledge of traditional Indigenous burning practices to inform regional bushfire management. *Local environment*, 24(8), 727-745.
- *Fulé, P. Z., Ramos-Gómez, M., Cortés-Montaño, C., & Miller, A. M. (2011). Fire regime in a Mexican forest under indigenous resource management. *Ecological Applications*, 21(3), 764-775.

Results of the first phase



INTEGRAL TRADITIONAL KNOWLEDGE				
Fire use practice	Agricultural and livestock ^{4,5,7,8,9,14}	Indigenous forest management ^{1,2,3,6,7,11,15}	Ritual, cultural and domestic uses ^{3,5,7,11,15}	Community prevention and response ^{3,12,13}
Objective	Prepare soils, reduce weeds, promote pasture regrowth.	Regulate fires and maintain biodiversity.	Ceremonial, medicinal, path and canal cleaning, product collection.	Reduce fire risks and control unplanned fires.
Activities	Pile burning, clearing with controlled fire lines, use of ashes.	Cultural burnings in forests; mosaics adapted to local context.	Burning in ritual and domestic spaces.	Local brigades, monitoring, record keeping.
Intensity/frequency	Low to moderate intensity / Seasonal or variable depending on drought.	Low to moderate intensity / Historically frequent.	Low intensity / Sporadic, tied to the cultural calendar.	Variable intensity depending on fire incidence / Higher in the dry season.
Supervision	Family or community-based.	Community-based; intergenerational transmission.	Family or ritual leaders.	Ejido members and organized communities.
Regulatory framework	Restrictive policies; burning prohibited in protected natural areas. Community integration recommended.	Partially recognized; lacks full integration.	Poorly recognized; risk of criminalization.	Insufficient and uneven regulations; legislative strengthening and community-based management recommended

Bardsley et al., 2019¹; Hoffman, 2021²; Long et al., 2021³; Smith et al., 2022⁴; Smith et al., 2024⁵; Vazquez-Varela et al., 2022⁶; Martínez-Torres et al., 2016⁷; Monzón-Alvarado & Keys, 2017⁸; Díaz-Toribio & Martorell, 2022⁹; Fulé et al., 2011¹⁰; Rodríguez-Trejo et al., 2011¹¹; Rodríguez-Cárdenas & Santacruz de León, 2021¹²; Bassaber-Zúñiga et al., 2024¹³; Ponce-Calderón et al., 2020¹⁴; Ponce-Calderón et al., 2022¹⁵.

I designed an interview for members of communities in the Meseta Purépecha who participate in cultural fire practices. Explored topics: (i) history, (ii) organization, (iii) objectives, (iv) effects on the landscape, (v) regulatory norms, (vi) community participation, and (vii) current challenges.

Technological reflections

Tool	Advantages	Aspects to consider
Cloud-based Geographic Information Systems (e.g., Google Earth Engine)	- Analysis of large volumes of satellite data. - Historical monitoring of fire scars and vegetation. - Does not require high-performance computing equipment	- Stable internet connection required. - Moderate learning curve.
Literature review platforms (e.g., CONSENSUS)	- Locates relevant articles. - Efficient filtering and organization of references. - Saves time.	- May exclude gray/local literature (not a systematic review). - Dependent on algorithm
Qualitative analysis software	- Codes and organizes interviews; identifies cultural and ecological patterns. - Provides rigor and reproducibility.	- Requires training. - Risk of losing cultural nuances.
Interview and automatic transcription applications	- Saves transcription time. - Facilitates online interviews and with external experts.	- Connectivity issues in rural communities. - Errors with accents and linguistic variants.

*Hoffman, K. M., Davis, E. L., Wickham, S. B., Schang, K., Johnson, A., Larking, T., ... & Trant, A. J. (2021). Conservation of Earth's biodiversity is embedded in Indigenous fire stewardship. *Proceedings of the National Academy of Sciences*, 118(32), e2105073118.

*Ponce-Calderon, L., Limón-Aguirre, F., Rodríguez, I., Rodríguez-Trejo, D. A., Bilbao, B., Álvarez-Gorriilo, G., & Villanueva-Díaz, J. (2022). Fire management in pyrobiocultural landscapes, Chiapas, Mexico. *Tropical Forest Issues*, 61, 53-59.

*Rodríguez-Trejo, D. A., Martínez-Hernández, P. A., Ortiz-Contla, H., Chavarría-Sánchez, M. R., & Hernández-Santiago, F. (2011). The present status of fire ecology, traditional use of fire, and fire management in Mexico and Central America. *Fire Ecology*, 7(1), 40-56.

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